Expanding Double Brackets

AQA KS3 · Year 8 · Foundation

Name: ___________________ Class: _________ Date: _________

What you need to be able to do

Expand products of two binomials (double brackets) using the grid method or FOIL. Collect like terms to write a simplified quadratic expression.

Common mistakes to avoid

- **Forgetting the last term: you must multiply the constant in the first bracket by the constant in the second.**
- **Sign errors with negatives: remember that a negative times a negative gives a positive.**
- **Writing x + x instead of 2x for the middle term when collecting like terms.**
- **Forgetting that $x \times x = x^2$, not 2x.**

Method steps

1. **Step 1: Draw a grid with the terms of the first bracket down the left and the terms of the second bracket along the top.**

2. Step 2: Fill in each cell by multiplying the row header by the column header.
3. Step 3: Write out all four products in a line.
4. Step 4: Collect like terms (the two middle terms will usually be alike).
5. Step 5: Check by substituting a simple value such as $x = 10$.

Worked example

Expand and simplify: $(x + 2)(x + 3)$

Step 1 — Draw the grid: columns are x and $+3$, rows are x and $+2$.

Step 2 — Fill in: $x \cdot x = x^2$, $x \cdot 3 = 3x$, $2 \cdot x = 2x$, $2 \cdot 3 = 6$

Step 3 — Write out: $x^2 + 3x + 2x + 6$

Step 4 — Collect like terms: $3x + 2x = 5x$

Answer: $x^2 + 5x + 6$

Check ($x = 10$): $(12)(13) = 156$ and $100 + 50 + 6 = 156$

[1 mark]

Expand and simplify.

$(x + 1)(x + 2)$

How to start

1. Draw a 2×2 grid. Put x and +1 down the left side, x and +2 along the top.
2. Fill in each cell: $x \cdot x$, $x \cdot 2$, $1 \cdot x$, $1 \cdot 2$.
3. Write all four terms, then collect the like terms together.

Answer: ____________________

____

Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

[1 mark] ●○○○○

Expand and simplify.

$(x + 3)(x + 4)$

How to start

1. Draw a 2×2 grid. Put x and $+3$ down the left side, x and $+4$ along the top.
2. Fill in each cell by multiplying the row term by the column term.
3. Add up the four products and collect like terms.

Answer: ____________________

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Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

[2 marks]

Expand and simplify.

$$(x + 5)(x - 2)$$

Answer: ____________________

____

Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

[2 marks] ●●○○○

Expand and simplify.

$(x - 3)(x + 4)$

Answer: ____________________

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Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

[2 marks] ●●●○○

Expand and simplify.

$$(x - 2)(x - 5)$$

Answer: ____________________

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Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

[3 marks] ●●●○○

Expand and simplify.

$(2x + 1)(x + 3)$

Answer: ______________________

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Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

[3 marks]

Expand and simplify.

$(3x - 2)(x + 4)$

Answer: ______________________

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Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

[3 marks] ●●●●●

Expand and simplify.

$(2x + 3)(3x - 1)$

Answer: ______________________

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Remember the method:

1 Draw the grid   2 Multiply each pair   3 Write all four products

4 Collect like terms   5 Check by substituting

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

Which question did you find the hardest, and why?

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One thing I now understand about expanding double brackets:

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One thing I will practise before the next lesson:

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Teacher key

Answers

1. Q1: $x^2 + 3x + 2$ [(x+1)(x+2): $x^2 + 2x + x + 2 = x^2 + 3x + 2$]
2. Q2: $x^2 + 7x + 12$ [(x+3)(x+4): $x^2 + 4x + 3x + 12 = x^2 + 7x + 12$]
3. Q3: $x^2 + 3x - 10$ [(x+5)(x-2): $x^2 - 2x + 5x - 10 = x^2 + 3x - 10$]
4. Q4: $x^2 + x - 12$ [(x-3)(x+4): $x^2 + 4x - 3x - 12 = x^2 + x - 12$]
5. Q5: $x^2 - 7x + 10$ [(x-2)(x-5): $x^2 - 5x - 2x + 10 = x^2 - 7x + 10$]
6. Q6: $2x^2 + 7x + 3$ [(2x+1)(x+3): $2x^2 + 6x + x + 3 = 2x^2 + 7x + 3$]
7. Q7: $3x^2 + 10x - 8$ [(3x-2)(x+4): $3x^2 + 12x - 2x - 8 = 3x^2 + 10x - 8$]
8. Q8: $6x^2 + 7x - 3$ [(2x+3)(3x-1): $6x^2 - 2x + 9x - 3 = 6x^2 + 7x - 3$]